1. Apply data cleaning techniques on any dataset (e.g., Paper Reviews dataset in UCI repository). Techniques may include handling missing values, outliers and inconsistent values. A set of validation rules can be prepared based on the dataset and validations can be performed.

In [34]:

```
import pandas as pd
import json

with open('reviews.json', 'r', encoding='utf-8') as f:
    data = json.load(f)

df = pd.json_normalize(
    data['paper'],
    record_path=['review'],
    meta=['id', 'preliminary_decision'],
    record_prefix='review_',
    meta_prefix='paper_'
)

print(df.head())
```

```
  review_confidence review_evaluation  review_id review_lan  \
0                 4                 1          1         es
1                 4                 1          2         es
2                 5                 1          3         es
3                 4                 2          1         es
4                 4                 2          2         es

  review_orientation review_remarks  \
0                  0
1                  1
2                  1
3                  1
4                  0

                                         review_text review_timespan paper_id  \
0  - El artículo aborda un problema contingente y...      2010-07-05        1
1  El artículo presenta recomendaciones prácticas...      2010-07-05        1
2  - El tema es muy interesante y puede ser de mu...      2010-07-05        1
3  Se explica en forma ordenada y didáctica una e...      2010-07-05        2
4                                                         2010-07-05        2

  paper_preliminary_decision
0                     accept
1                     accept
2                     accept
3                     accept
4                     accept
```

In [35]:

```
# ========= handling missing values =========

print(df.isnull().sum())
```

```python
df['review_confidence'] = pd.to_numeric(df['review_confidence'], errors='coerce')
df['review_evaluation'] = pd.to_numeric(df['review_evaluation'], errors='coerce')

df.fillna({'review_confidence': df['review_confidence'].mean(), 'review_evaluation'

# df['review_remarks'].fillna('No remarks', inplace=True)
# df['review_text'].fillna('No text provided', inplace=True)

df.fillna({'review_remarks': '', 'review_text': ''}, inplace=True)
```

```
review_confidence           2
review_evaluation           0
review_id                   0
review_lan                  0
review_orientation          0
review_remarks              0
review_text                 0
review_timespan             0
paper_id                    0
paper_preliminary_decision  0
dtype: int64
```

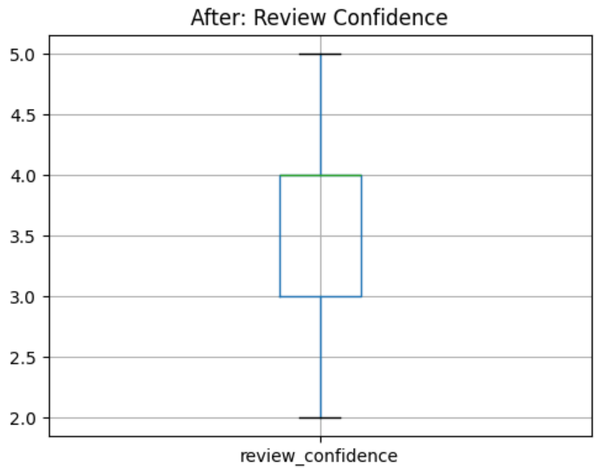
In [36]:
```python
# ========= handling inconsistent data =========

df['review_lan'] = df['review_lan'].str.strip().str.lower()
# df['review_lan'].replace({'spa': 'es', 'en-us': 'en'}, inplace=True)
df.replace({'review_lan': {'spa': 'es', 'en-us': 'en'}}, inplace=True)

df['paper_preliminary_decision'] = df['paper_preliminary_decision'].str.strip().str
df.loc[~df['paper_preliminary_decision'].isin(['accept', 'reject']), 'paper_prelimi
```

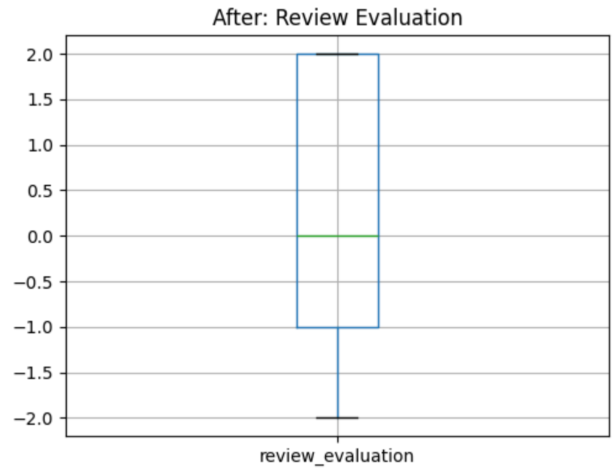
In [37]:
```python
# Visualize outliers before handling
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
df.boxplot(column=['review_confidence'])
plt.title('Before: Review Confidence')

plt.subplot(1, 2, 2)
df.boxplot(column=['review_evaluation'])
plt.title('Before: Review Evaluation')
plt.tight_layout()
plt.show()
```

In [38]:
```python
# ========= handling outliers using IQR method =========

for col in ['review_confidence', 'review_evaluation']:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    print(f"Outliers found in {col}: {len(outliers)}")

    # Remove outliers
    df = df[(df[col] >= lower_bound) & (df[col] <= upper_bound)]

print(f"\nShape of DataFrame after removing outliers: {df.shape}")
```

```
Outliers found in review_confidence: 7
Outliers found in review_evaluation: 0

Shape of DataFrame after removing outliers: (398, 10)
```

In [39]:
```python
# Visualize data after handling outliers
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
df.boxplot(column=['review_confidence'])
plt.title('After: Review Confidence')

plt.subplot(1, 2, 2)
df.boxplot(column=['review_evaluation'])
plt.title('After: Review Evaluation')
plt.tight_layout()
plt.show()
```





Validation Rules:

Rule	Field	Expected Range / Type
R1	review_confidence	1–5
R2	review_evaluation	-2–2
R3	review_lan	one of ['en', 'es', 'fr']
R4	paper_preliminary_decision	['accept', 'reject']
R5	review_text	not empty

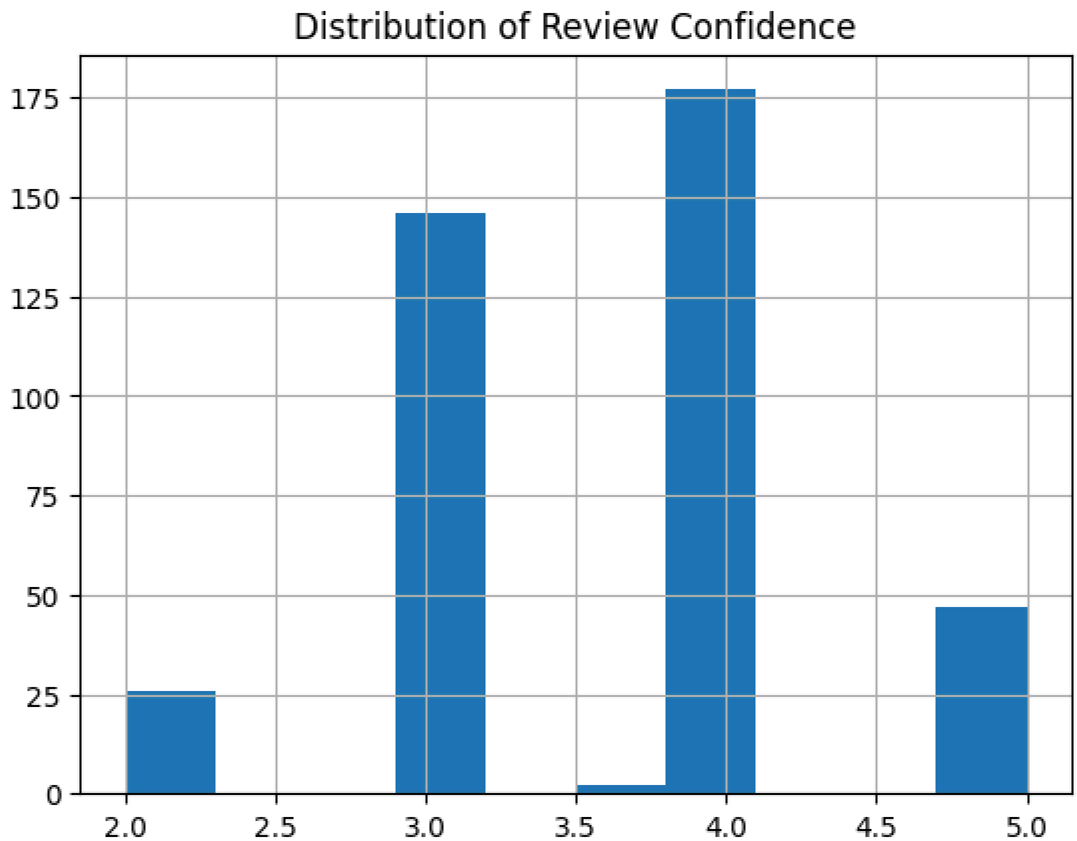
In [42]:
```
# ========= applying validation rules =========

rule1 = df["review_confidence"].between(1, 5)
rule2 = df["review_evaluation"].between(-2, 2)
rule3 = df["review_lan"].isin(["en", "es", "fr"])
rule4 = df["paper_preliminary_decision"].isin(["accept", "reject"])
rule5 = df["review_text"].str.strip().ne("")

Rules = pd.DataFrame({
    "Rule 1 (Confidence 1-5)": rule1,
    "Rule 2 (Evaluation -2-2)": rule2,
    "Rule 3 (Language Valid)": rule3,
    "Rule 4 (Decision Valid)": rule4,
    "Rule 5 (Text Non-empty)": rule5
})

print("Rule Violation Summary:")
for col in Rules.columns:
    total = len(Rules)
    passed = Rules[col].sum()
    failed = total - passed
    print(f"{col}: {failed} violations out of {total}")
```

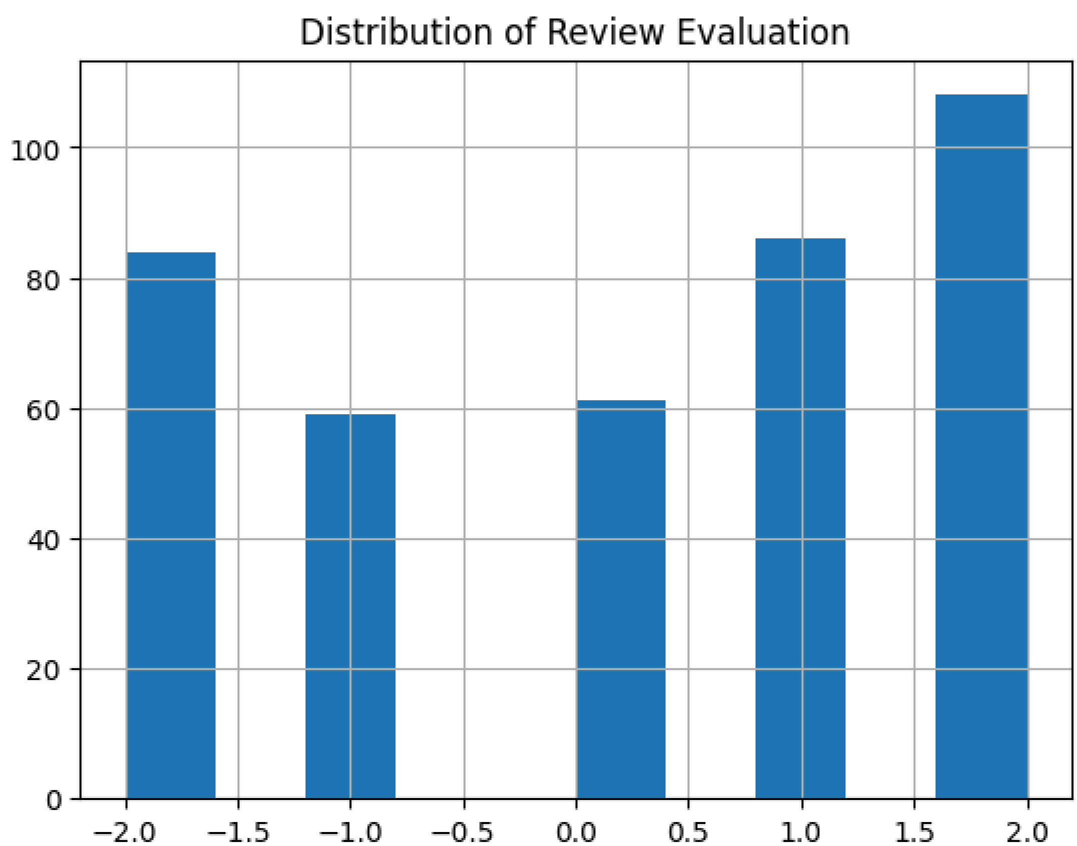
```
Rule Violation Summary:
Rule 1 (Confidence 1-5): 0 violations out of 398
Rule 2 (Evaluation -2-2): 0 violations out of 398
Rule 3 (Language Valid): 0 violations out of 398
Rule 4 (Decision Valid): 21 violations out of 398
Rule 5 (Text Non-empty): 6 violations out of 398
```

In [41]:

```python
# ========= final cleaned data =========

print(df.describe(include='all'))

# Visualize numeric distributions
import matplotlib.pyplot as plt
df['review_confidence'].hist()
plt.title('Distribution of Review Confidence')
plt.show()

df['review_evaluation'].hist()
plt.title('Distribution of Review Evaluation')
plt.show()

# save cleaned data
df.to_csv("cleaned_paper_reviews.csv", index=False)
```

```
       review_confidence  review_evaluation   review_id review_lan  \
count          398.000000         398.000000  398.000000        398
unique                NaN                NaN         NaN          2
top                   NaN                NaN         NaN         es
freq                  NaN                NaN         NaN        381
mean             3.618458           0.188442    1.819095        NaN
std              0.776587           1.504709    0.816972        NaN
min              2.000000          -2.000000    1.000000        NaN
25%              3.000000          -1.000000    1.000000        NaN
50%              4.000000           0.000000    2.000000        NaN
75%              4.000000           2.000000    2.000000        NaN
max              5.000000           2.000000    4.000000        NaN

      review_orientation review_remarks review_text review_timespan  \
count                398            398         398             398
unique                 5            108         393               4
top                   -1                                 2014-07-05
freq                 139            288           6             124
mean                 NaN            NaN         NaN             NaN
std                  NaN            NaN         NaN             NaN
min                  NaN            NaN         NaN             NaN
25%                  NaN            NaN         NaN             NaN
50%                  NaN            NaN         NaN             NaN
75%                  NaN            NaN         NaN             NaN
max                  NaN            NaN         NaN             NaN

        paper_id paper_preliminary_decision
count      398.0                        377
unique     168.0                          2
top        112.0                     accept
freq         4.0                        257
mean         NaN                        NaN
std          NaN                        NaN
min          NaN                        NaN
25%          NaN                        NaN
50%          NaN                        NaN
75%          NaN                        NaN
max          NaN                        NaN
```


Distribution of Review Confidence
175
150
125
100
75
50
25
0
2.0
2.5
3.0
3.5
4.0
4.5
5.0


Distribution of Review Evaluation
100
80
60
40
20
0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0